

# Break the Memory Wall.

Serve 70B LLMs on a single DGX Spark with **24× faster** time-to-first-token.

**24** ×

Faster TTFT

187 s → 8 s

**11** ×

Query throughput

3,705 s → 339 s

**100** %

Local-NVMe latency parity

+ external scale

vLLM + LMCache + SupremeRAID™ AE / NVMe-oF · RoCEv2 / NVIDIA DGX Spark (GB10) / Qwen2.5-72B-Instruct (AWQ)

# Prefill is expensive. Local storage runs out.

Every LLM request reprocesses the full context to build its KV cache before a single token is generated. For long-context RAG and multi-turn agents, the same KV state is computed over and over — burning GPU time on work already done.

## Capacity ceiling

DGX Spark ships with just 4 TB local NVMe. Production multi-tenant workloads overrun that within hours.

## Latency tax

The retrieve path must beat re-prefilling on the GPU — or offloading is pointless.

WITHOUT KV CACHE OFFLOADING

**187** <sup>s</sup> Time to first token

**3,705** <sup>s</sup> To clear one query round

Qwen2.5-72B-AWQ · 60 docs · 16 in-flight

# One integrated stack, collapsed onto a single GB10.

**L4** **vLLM** 0.21.0 SERVING

↑ KV offload / retrieve

**L3** **LMCache** 0.4.5 CACHE TIER

↑ block I/O

**L2** **SupremeRAID™ AE** RAID ON GPU (HOST)

↑ RDMA · kernel bypass

**L1** **NVMe-oF / RoCEv2** 100 GB/S FABRIC

THE ENABLER

**NVIDIA CUDA MPS.** RAID parity, NVMe-oF servicing, KV offload writes, and model execution all share one GB10's SMs and unified memory under a single arbitrator.

- **vLLM 0.21.0**  
OpenAI-compatible serving with continuous batching and native prefix caching.

- **LMCache 0.4.5**  
Transparently offloads and retrieves vLLM KV caches to a disk tier — no model changes.

- **SupremeRAID™ AE**  
Host-side GPU-accelerated NVMe RAID across the 4 NVMe-oF subsystems. Tested at RAID 5 (3+1); supports 0/1/5/6/10.

- **NVMe-oF / RoCEv2**  
Lossless 100 Gb/s RDMA fabric (MTU 9000) — kernel-bypass DMA straight to the GP Spark NVMe-oF target.

# A petaFLOP on your desk.

MSI EdgeXpert MS-C931 — an NVIDIA DGX™ Spark–class node in a 1.2 kg, 1.19 L chassis.

COMPUTE NODE

## MSI EdgeXpert MS-C931

NVIDIA® GB10 Grace Blackwell · DGX OS



Same **GB10 Grace Blackwell Superchip** at the core of NVIDIA DGX™ Spark — **1 petaFLOP** of FP4 tensor performance, 128 GB unified memory, in a desktop form factor.

|              |                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------|
| SUPERCHIP    | NVIDIA® GB10 Grace Blackwell<br>Blackwell GPU + 20-core ARM CPU (10× X925 + 10× A725) via NVLink-C2C          |
| TENSOR PERF  | <b>1,000 AI TOPS</b> · FP4 (sparse)<br>1 petaFLOP — runs LLMs up to 200 B params locally, 405 B paired        |
| MEMORY       | <b>128 GB</b> LPDDR5X unified · <b>273 GB/s</b><br>CPU + GPU coherent — no host↔device copies                 |
| STORAGE      | <b>1TB or 4 TB</b> NVMe M.2 Gen5 · self-encrypting                                                            |
| FABRIC       | NVIDIA® ConnectX-7 SmartNIC · <b>RoCEv2</b> · <b>200 Gb/s</b><br>The gateway to NVMe-oF                       |
| CHASSIS & OS | 151 × 151 × 52 mm · 1.2 kg · NVIDIA DGX™ OS<br>Role: runs vLLM + LMCache; offloads KV to GP Spark over RoCEv2 |

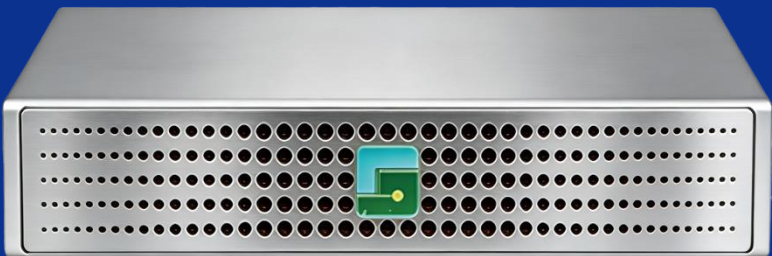
# RAID-protected flash, far past the 4 TB local ceiling.

GP Spark — the NVMe-oF target appliance behind SupremeRAID™ AE. The KV cache tier the GB10 was waiting for.

STORAGE TARGET

## GP Spark

NVMe-oF target · RoCEv2 · GPU Direct Storage



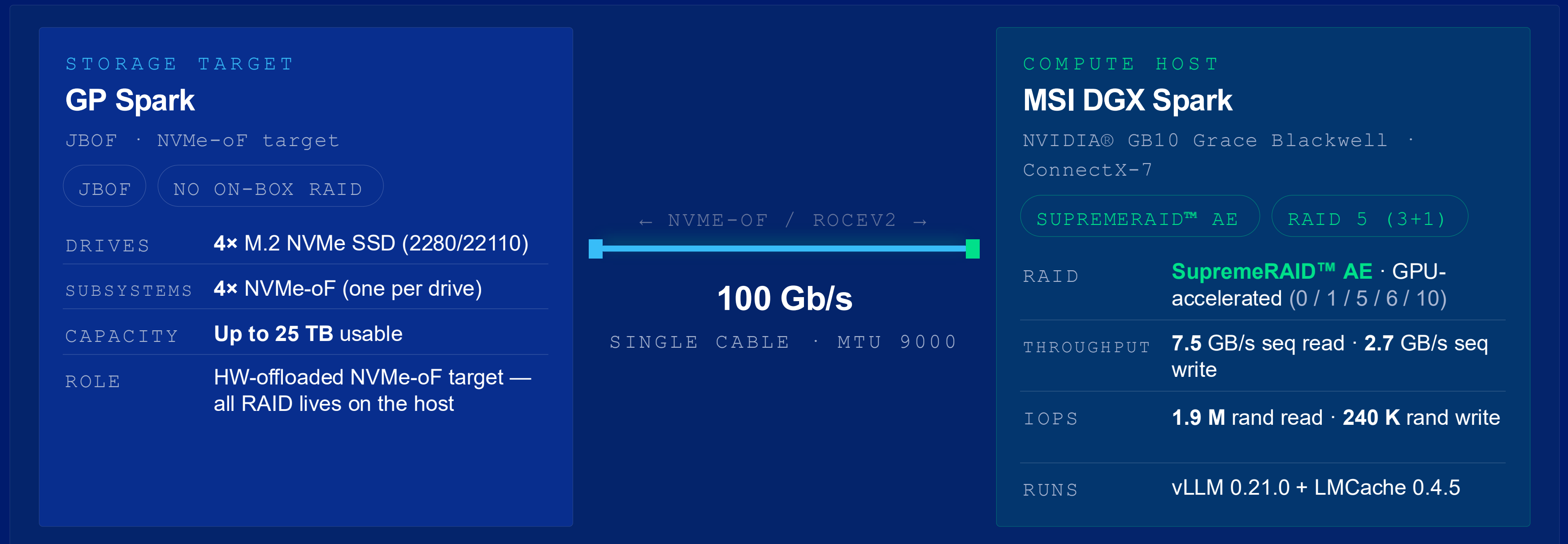
A compact NVMe-oF target exposing **4× M.2 NVMe SSDs (2280/22110)** as **4 NVMe-oF subsystems**, **RoCEv2+NVMe-oF offloaded in hardware** . SupremeRAID™ AE on the host stripes them into one RAID-protected virtual drive — **no CPU cycles spent on parity**.

|          |                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DRIVES   | <b>4× M.2 NVMe SSD (2280/22110)</b><br>Each exposed as its own NVMe-oF subsystem (4 subsystems total)                                                                   |
| CAPACITY | <b>Up to 25 TB usable</b><br>vs 4 TB local on a single DGX Spark                                                                                                        |
| TOPOLOGY | Smart <b>JBOF</b> · no on-box RAID controller<br>RoCEv2 + NVMe-oF offloaded in hardware; RAID & parity on the host's SupremeRAID™ AE                                    |
| FABRIC   | NVMe-oF / RoCEv2 · single-port 100 GbE (≈ 12 GB/s line rate)<br>Role: dedicated KV cache tier for one DGX Spark — re-hydrates prefix in ≈ 8 s · < 20 μs media · < 100 W |



# How the boxes connect.

A compact NVMe-oF target on one end, the GB10 host running SupremeRAID™ AE on the other — one cable between them.



# What the numbers say.

Qwen2.5-72B-AWQ · 60 docs · 16 in-flight · LMCache + SupremeRAID AE vs no cache



# 24×

**Faster time-to-first-token**

187 s → 8 s



# 11×

**Faster query throughput**

3,705 s → 339 s



# 66.2%

**External KV cache hit rate**

served from storage, not  
GPU



# ≈ 8<sub>s</sub>

**Retrieve latency**

near-constant across  
model sizes

# From 3 minutes to 8 seconds.

Time-to-first-token on Qwen2.5-72B-AWQ, query round

BEFORE

No KV cache

# 187

seconds

Every prompt is fully reprocessed on the GPU. Every multi-turn agent step pays the prefill bill again.

# 24×

FASTER

AFTER

LMCache + SupremeRAID AE / NVMe-oF

# 8

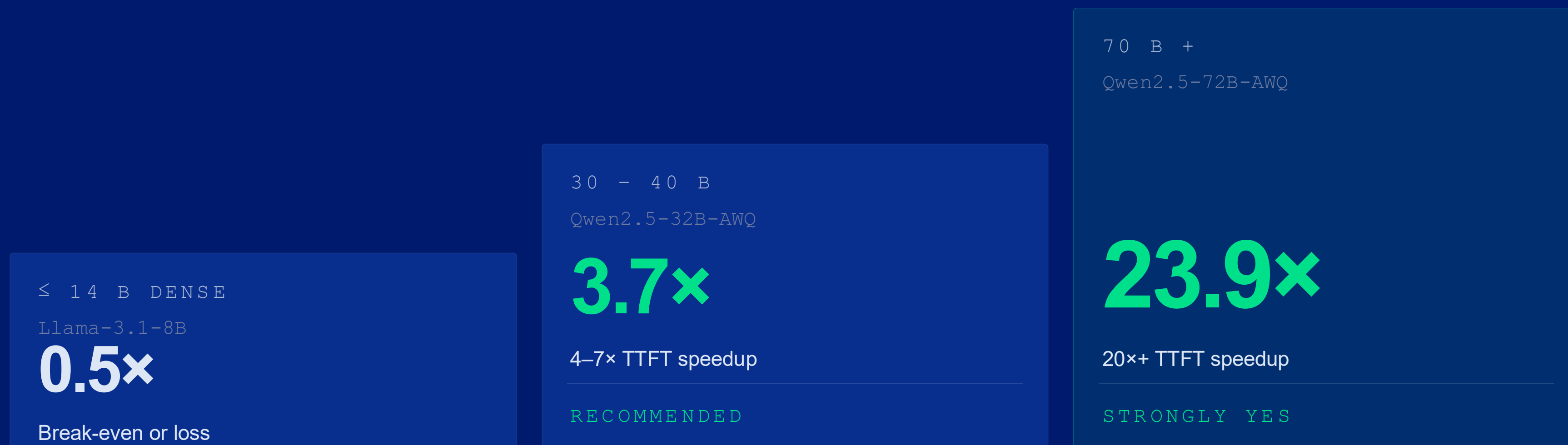
seconds

Prefix KV is restored from RAID-protected flash over a kernel-bypass fabric — no re-prefill on the GPU.



# The bigger the model, the bigger the win.

Each retrieve costs roughly the same (~8 s) — so the more compute a cold prefill would burn, the more LMCache saves. Measured speedups, three model classes:



KEEP KV IN GPU

# Local-NVMe speed. None of the limits.

LATENCY PARITY

QUERY TTFT · 32B-AWQ

Local ext4 NVMe

98<sub>s</sub>SupremeRAID AE /  
NVMe-oF104<sub>s</sub>

Δ 6%

WITHIN RUN-TO-RUN VARIANCE



## Scale past 4 TB

~25 TB available today and scaling further — long-context, multi-tenant caches that a single host SSD can't hold.



## Dedicated, not shared

One GP Spark pairs with one DGX Spark over a single 100 GbE link. RAID and parity run host-side on SupremeRAID™ AE — a clean, single-owner KV tier with no contention.



## RAID protection

A single SSD failure won't wipe a warmed corpus. RAID 5/6/10 means the cache survives — no full re-prefill.

# GPU-accelerated RAID, measured.

Raw fio numbers on the SupremeRAID AE + NVMe-oF tier — and headroom still on the table.

|                                                           |                                                            |                                                         |                                                          |
|-----------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| <div>SEQUENTIAL READ</div> <div>7.5</div> <div>GB/s</div> | <div>SEQUENTIAL WRITE</div> <div>2.7</div> <div>GB/s</div> | <div>RANDOM READ</div> <div>1.9 M</div> <div>IOPS</div> | <div>RANDOM WRITE</div> <div>240 K</div> <div>IOPS</div> |
|-----------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|

WARMUP PHASE

KV evicted → flash

Seq write2.7 GB/s

Random write240 K IOPS

Paid once

per prefix — amortized over every future query that hits it.

QUERY PHASE

KV retrieved ← flash

Seq read7.5 GB/s

Random read1.9 M IOPS

Paid on every reuse

— the hot path that turns 187 s TTFT into 8 s.

# Single-digit-second TTFT. No new GPU memory required.

For **prefix-reusing** workloads on **30B–70B models** — long-context RAG, multi-turn agents, repeated-prompt analytics — pairing **vLLM + LMCache** with **SupremeRAID™ AE** over **NVMe-oF** delivers single-digit-second TTFT on DGX Spark, with latency parity to local NVMe plus the capacity and RAID protection local storage can't deliver — from one compact, dedicated NVMe-oF appliance per node. Smaller models ( $\leq 14$  B) and prompt-diverse traffic see less or no benefit.

**24×**

Faster TTFT

**11×**

Higher throughput

**Δ 6%**

Latency parity vs local

**RAID 5/6/10**

Protected &amp; dedicated

**SupremeRAID™ AE**

GPU-ACCELERATED NVME RAID

BY GRAID TECHNOLOGY

DGX SPARK · GB10 · VLLM 0.21.0 · LMCACHE 0.4.5 · NVME-OF 100 GB/S